# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\frac{d(V_i)}{d(V_3)} = \lim_{i\tau\nearrow 0} \frac{\sum_{j=1}^{4} S_{j,i}\chi_{V_j}(-\frac{1}{\tau})}{\sum_{k=1}^{4} S_{k,3}\chi_k(-\frac{1}{\tau})} = \lim_{q\to 0} \frac{\sum_{j=1}^{4} S_{j,i}\chi_j(q)}{\sum_{k=1}^{4} S_{k,3}\chi_k(q)} = \frac{S_{3,i}}{S_{3,3}}.$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\left[ \partial_\mu \, \partial^\mu + \frac{V_p}{(p+1)!} \, \frac{\partial^2}{\partial\sigma_{\mu_1 \dots \mu_{p+1}} \, \partial\sigma^{\mu_1 \dots \mu_{p+1}}} - (p+1) \, M_0^2 \right] G \, ( \, x - x_0 \ , \sigma \, ) = \delta^D \, ( \, x - x_0 \, ) \ \delta \, ( \, \sigma \, ) \ \ .$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$\frac{\partial}{\partial r}\left[(r^2+A^2-2Mr)\frac{\partial G}{\partial r}\right] + \frac{1}{sin\theta}\frac{\partial}{\partial\theta}\left[sin\theta\frac{\partial G}{\partial\theta}\right] = -\delta(r-r_o)\delta(cos\theta - cos\theta_o)\delta(\varphi-\varphi_o)$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\frac{\partial\zeta^{(\lambda\mu\alpha)}}{\partial\zeta^{(\tau,\nu,\beta)}} = \psi^{(\lambda\mu\alpha)}_{(\tau,\nu,\beta)} = \frac{1}{2}\left(D^{(\lambda\mu\alpha)}_{(\tau,\nu,\beta)} + \omega^{(\rho,\omega,\gamma)}_{(\tau,\nu,\beta)} D^{(\lambda\mu\alpha)}_{(\rho,\omega,\gamma)}\right).$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\sup_{\mathcal{T}\geq 1} |\frac{1}{\mathcal{T}}\int_0^{\mathcal{T}} \tilde{G}(\mu^{N,M}(t), z^{N,M}(t))dt - \frac{1}{\mathcal{T}}\int_0^{\mathcal{T}} \tilde{G}(\bar{\mu}^{N,M}(t), \bar{z}^{N,M}(t))dt| \leq \kappa_3(\epsilon), \qquad \lim_{\epsilon\to 0}\kappa_3(\epsilon) = 0.$$